Yelamarthi Sai Krishna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.77
Sri Chaitanya Junior Kalasala, Kukatpally	Class XII (Board of Intermediate Education, Andhra Pradesh)	2012	91.50%
Silver Oaks High School	Class X (Board of Secondary Education, Andhra Pradesh)	2010	84.83%

WORK EXPERIENCE

Design Engineer, L&T Technology Services, Bengaluru | 2018-2021

- Design and development of automotive sheet-metal and plastic components.
- Applied DFM and DFA criteria to each design
- Worked with the CAE team to iterate designs against structural failure criteria

DOCTORAL RESEARCH

- Developed a semi-supervised deep learning framework for damage-mode classification in carbon fiber composite materials using acoustic emission waveforms.
- Built an unsupervised deep learning pipeline using time-frequency representation of acoustic emission waveforms to characterize the damage evolution of carbon fiber composite materials.
- Compared the damage behavior of carbon fiber composite materials with different laminate architectures using experiments, acoustic emission, and Abaqus-based progressive damage modelling.
- Developed a neural network-based technique for full-field deformation measurement.

JOURNAL PUBLICATIONS

- **Yelamarthi Sai Krishna**, Gangadharan Raju, Maunendra Sankar Desarkar, Damage characterization of CFRP composites under compression using acoustic emission time-frequency analysis and unsupervised deep learning, Mechanical Systems and Signal Processing, Volume 238, 2025, 113224, ISSN 0888-3270, https://doi.org/10.1016/j.ymssp.2025.113224.
- **Sai Krishna Y**, Raju G, Desarkar MS. Damage mode classification in CFRP laminates using convolutional autoencoder and convolutional neural network on acoustic emission waveforms. *Structural Health Monitoring*. 2026;25(2):702-724, doi:10.1177/14759217241298403.
- **Yelamarthi Sai Krishna**, P. Mahesh, Gangadharan Raju, M. Ramji, Maunendra Sankar Desarkar, Open-hole compression behavior of double-double laminates: A comparative study with quad laminates, Composite Structures, Volume 397, 2026, 120807, ISSN 0263-8223, https://doi.org/10.1016/j.compstruct.2026.120807.

CONFERENCE PRESENTATIONS

- Damage Classification in CFRP Composite Laminates using Acoustic Emissions and Machine Learning, *International Structural Integrity Conference and Exhibition (SICE 2022)*, 2022, Hyderabad, India.
- Neural network enhanced LaRC05 failure criteria for progressive damage modelling of CFRP, *ISAMPE National Conference on Composites (INCCOM 19)*, 2025, Pune, India.
- Mechanical characterization of carbon fiber-reinforced polymer composites using physics-informed neural networks, *European Conference on Composite Materials (ECCM22)*, 2026, Oslo, Norway.

SKILLS

Computational Mechanics: Finite Element Analysis, Cohesive Zone Modeling, Virtual Fields Methods

AI/ML: Physics-Informed Neural Networks, Convolutional Neural Networks, U-Net, Autoencoders, Data Efficient Learning, Image Segmentation

Experimental Techniques: Acoustic Emission, Digital Image Correlation, Full-Field Deformation Measurements

Programming & Libraries: Python, MATLAB, PyTorch, TensorFlow/Keras, NumPy, SciPy, Pandas, OpenCV, Scikit-learn, Matplotlib

Engineering Software: Abaqus, CATIA

AWARDS AND ACHIEVEMENTS

- **Best Project Award** - Awarded for Master's thesis work.
- **Best Oral Presentation Award** - Received at the *International Structural Integrity Conference and Exhibition 2022*

POSITIONS OF RESPONSIBILITY

- Design Engineer, L&T Technology Services, Bengaluru (2018-2021)
- Teaching Assistant, Machine Learning and Its Applications, Indian Institute of Technology Hyderabad (2025-2026)